SUPPLY CHAIN ANALYTICS REPORT

End-to-End Supply Chain Delivery Analysis

Why 6 in 10 orders arrive late, what it costs, and how to fix it

60.09%	$2.2M	100%	77%
of orders delivered late	profit carried by late orders	of First Class and Same Day orders late	accuracy of the late-order model

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Data	DataCo supply chain dataset: 172,765 orders, January 2015 to January 2018
Method	Python analysis (pandas, scikit-learn) in a Jupyter notebook
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1 Executive Summary

60.09% of orders (103,816 of 172,765) ship later than scheduled. These late orders carry $2.2M of profit, and the main cause is how shipping modes are set up, not where the orders go.

This report analyses 172,765 orders placed between January 2015 and January 2018 with a global e-commerce company that sells sporting goods, fitness equipment, apparel and footwear. It measures delivery performance, sizes the profit at stake, locates the bottlenecks and tests whether late orders can be predicted before they ship.

> **Headline findings**
>
> First Class and Same Day orders are late **100%** of the time and Second Class **79.8%**, against **39.8%** for Standard Class.
>
> Every top region sits between **60% and 66%** late, so the problem is company-wide and process-driven, not local.
>
> A Random Forest model flags late orders with **77% accuracy** and **83% precision**, so it can be piloted as an early-warning alert now.

Table 1. Key results

Measure	Result
Late delivery rate	60.09% (103,816 orders)
Early or on-time orders	39.91% (68,949 orders)
Profit carried by late orders	$2.2M
Worst shipping modes	First Class and Same Day: 100% late
Highest-delay region	Central Asia: 65.9% late
Loss-making orders	18.7% of orders
Late-order model (Random Forest)	77% accuracy; 0.83 precision on late orders

2 Business Context and Data

2.1 Business problem

The company runs end-to-end order fulfilment, including shipping and delivery, across many regions. Actual shipping times often miss the scheduled delivery window. This erodes customer trust, makes order profit unpredictable and stops the business from making reliable delivery promises at checkout.

2.2 Objectives

1. Measure delivery performance by region, shipping mode, customer segment, department and time.
2. Quantify how delivery delays relate to order profit.
3. Find the main bottlenecks and their root causes.
4. Build a model that flags orders likely to be late before they ship.
5. Recommend actions with measurable targets.

2.3 Data and definitions

Table 2. Dataset and key definitions

Item	Detail
Source	DataCo supply chain dataset, analysed in Python (Jupyter notebook)
Raw data	180,519 rows × 53 columns
After cleaning	172,765 orders × 22 columns
Removed	7,754 orders whose shipping was cancelled, and 31 columns (personal details, IDs and other unneeded fields)
Order period	1 January 2015 to 31 January 2018
Delay	Days from order to shipping, minus the scheduled shipping days
Late order	An order with a delay above 0 days
Profit	Order Profit Per Order

The predictive model in Section 8 uses the dataset's own Late_delivery_risk flag as its target, which marks 57.3% of orders as late. All other late rates in this report use the delay definition above.

3 Key Performance Indicators

Three in five orders are late, but 90% of those are at most 3 days late.

172,765 Total orders	**103,816** Late deliveries	**68,949** Early or on-time orders	**60.09%** Late delivery rate
$7.5M Profit from profitable orders	**$2.2M** Profit carried by late orders	**$22.03** Average profit per order	**3 days** 90th-percentile late delay
3.50 days Average actual time	**2.93 days** Average scheduled time	**+0.57 days** Average delay	**18.7%** Loss-making orders

- **Lateness is the norm.** At 60.09%, a late delivery is the typical customer experience, not an exception.
- **Delays are short but systematic.** Orders take 3.50 days on average against 2.93 scheduled, and 90% of late orders are at most 3 days late. That pattern points to planning and process, not breakdowns.
- **Profit is exposed.** Late orders carry $2.2M of profit.

4 Profitability Analysis

4.1 Order profitability

Four in five orders make money, but losses cancel out about half of gross profit.

Table 3. Orders by profitability

Category	Orders	Share of orders
Profitable	139,354	80.7%
Loss-making	32,295	18.7%
Break-even	1,116	0.6%
Total	**172,765**	**100.0%**

Profitable orders earn $7.5M. Net profit across all orders is about $3.8M, so loss-making orders take away roughly $3.7M.

4.2 Delivery delay and profit

31.0% of orders arrive exactly 1 day late, the largest single group.

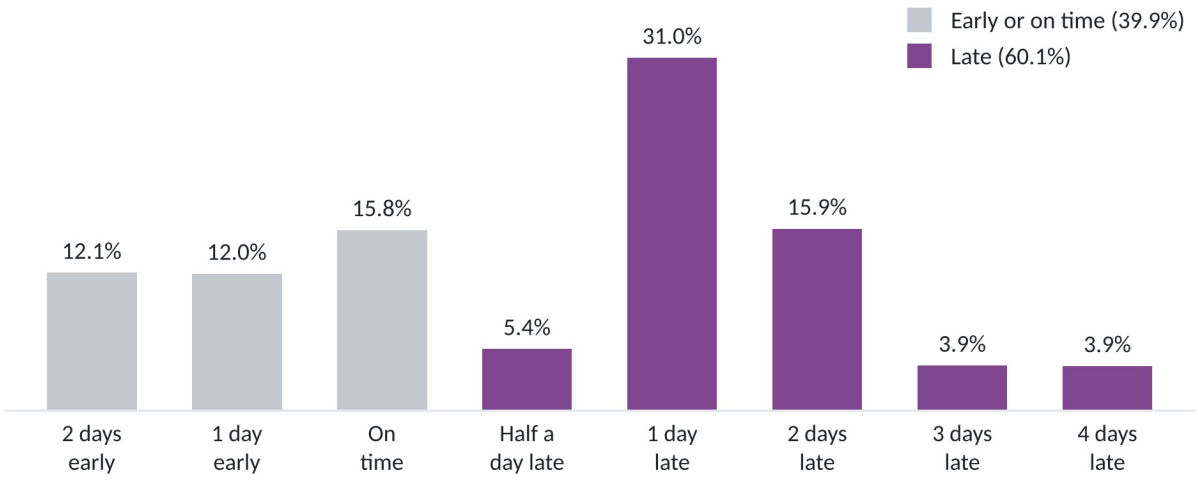


Figure 1. Share of orders by delivery delay (172,765 orders)

Table 4. Profit by delivery delay

Delay	Orders	Average profit per order	Total profit
2 days early	20,873	$23.36	$487.6K
1 day early	20,719	$21.60	$447.6K
On time	27,357	$22.71	$621.2K
Half a day late	9,293	$20.90	$194.2K
1 day late	53,503	$22.33	$1,194.9K
2 days late	27,551	$21.13	$582.1K
3 days late	6,772	$20.03	$135.7K
4 days late	6,697	$21.37	$143.1K

- **Small margin gap.** Early or on-time orders average $22.57 profit; late orders average $21.67. The lowest average is $20.03, at 3 days late.
- **The exposure is volume, not margin.** Because the per-order gap is under $1, the main financial risk is the $2.2M of profit that sits on late orders.
- **The half-day group is Same Day shipping.** Its 9,293 orders match the 9,293 Same Day orders exactly: they are scheduled for 0 days and ship after about half a day, so every one counts as late.

5 Bottleneck Detection

Of the six dimensions compared, only shipping mode moves the late rate a lot: from 39.8% for Standard Class to 100% for First Class and Same Day. Region, customer segment, order status, payment type and department all stay within a few points of the 60.1% average.

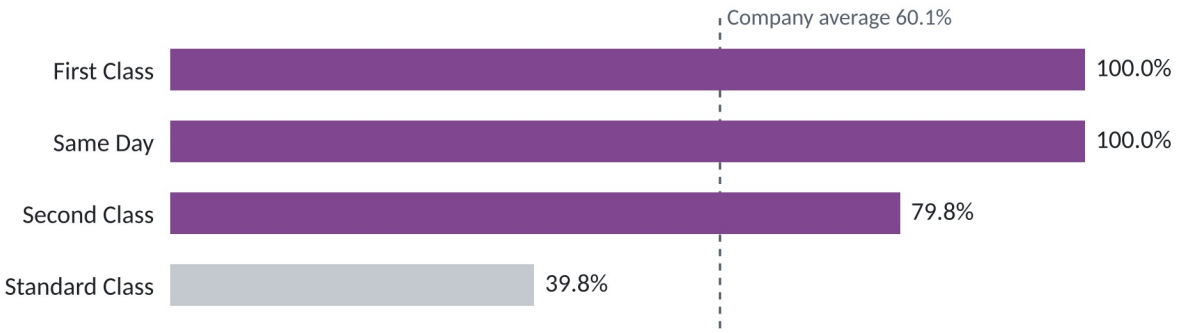


Figure 2. Late rate by shipping mode

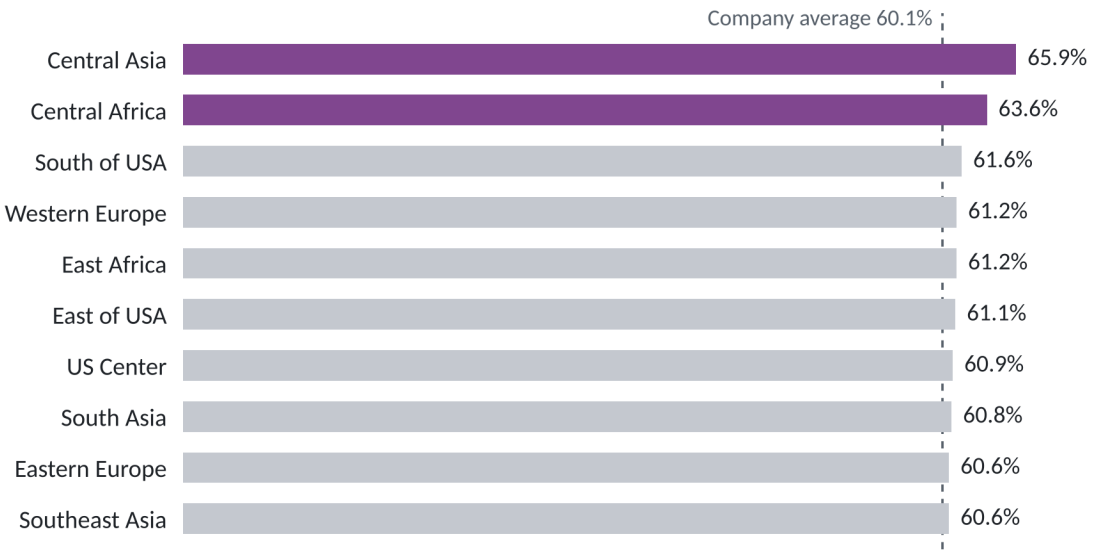


Figure 3. Late rate by order region (notebook's top 10)

Table 5. Late rate range in the other dimensions

Dimension	Highest	Lowest shown	Spread
Customer segment	Home Office 60.4%	Corporate 59.8%	0.6 pts
Order status	PENDING 60.8%	ON_HOLD 58.3%	2.5 pts
Payment type	TRANSFER, PAYMENT 60.3%	CASH 59.5%	0.8 pts
Department (top 10)	Pet Shop 64.6%	Footwear, Fan Shop 60.0%	4.6 pts
Region (top 10)	Central Asia 65.9%	Eastern Europe, Southeast Asia 60.6%	5.3 pts

Region and department show the notebook's top 10 only, so their true lowest values may be lower.

- **Shipping mode is the main lever.** First Class orders (26,513) and Same Day orders (9,293) are late every time and Second Class 79.8% of the time; Standard Class does best at 39.8%.
- **The problem is company-wide.** All top-10 regions are 60.6% to 65.9% late, and the three customer segments are within 0.6 points of each other.
- **Department differences are small.** Pet Shop leads at 64.6%; the rest of the top 10 sit between 60.0% and 61.1%.

6 Root Cause Analysis: Central Africa

In Central Africa (63.6% late, the second-highest region), shipping mode again leads, and payment holds appear as a second bottleneck.

The notebook ranks every level of five drivers within the region: shipping mode, customer segment, department, payment type and order status. Figure 4 shows the ten levels with the highest late rates.

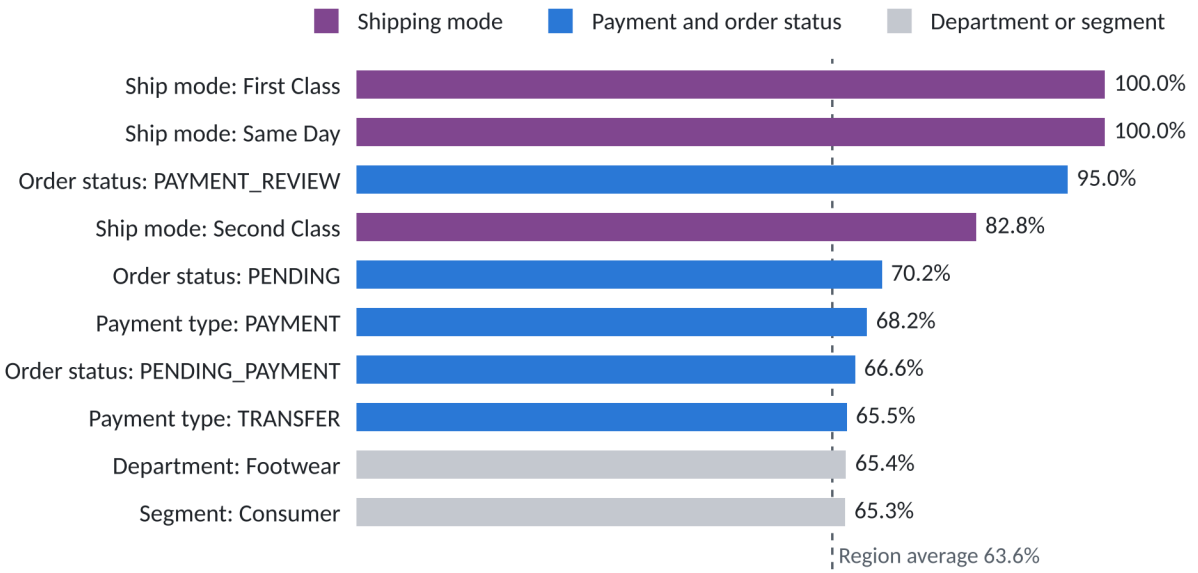


Figure 4. Top 10 drivers of late delivery in Central Africa

- **Shipping mode.** First Class and Same Day are 100% late in the region and Second Class 82.8%.
- **Payment holds.** 95.0% of PAYMENT_REVIEW orders and 70.2% of PENDING orders ship late, against 63.6% for the region. Company-wide, PAYMENT_REVIEW orders are only 60.2% late, so this is a regional problem.

> **Caveats**
>
> PAYMENT_REVIEW covers only 1,893 orders company-wide (1.1%), so the Central Africa figure rests on a small number of orders. Confirm the count before acting on it.
>
> Central Asia has the highest late rate (65.9%), but the notebook's deep dive covers Central Africa. Central Asia is the natural next deep dive.

7 Time-Based Delay Patterns

Timing barely matters. The monthly late rate stays between 59.2% and 60.7%, and the hourly rate between 57.7% and 62.4%.

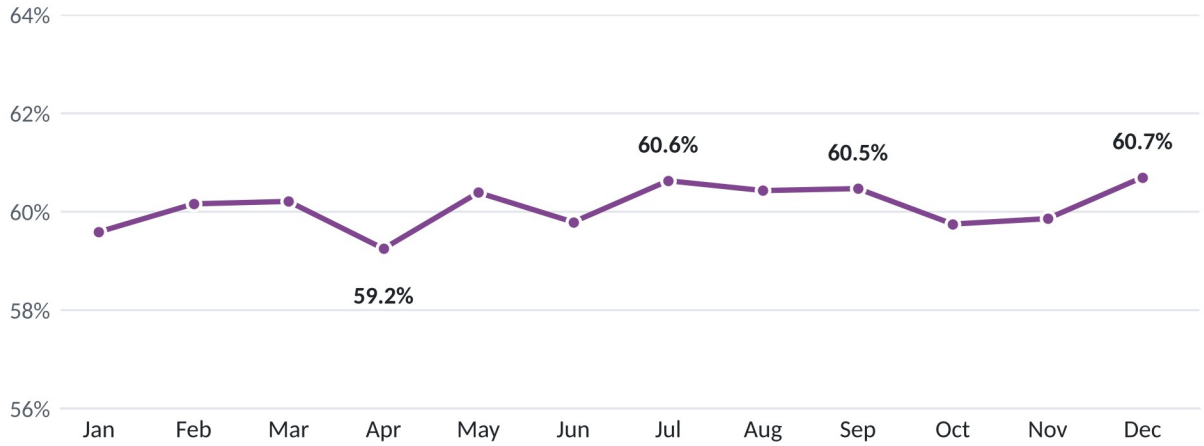


Figure 5. Late rate by order month (all years combined)

Labelled values for December, July and September are the notebook's annotations; April and the other months are read from its chart.

Table 6. Highest and lowest hours of the day

Hour the order was placed	Late rate	Note
12:00	62.4%	Highest
20:00	61.7%	Second highest
01:00	61.4%	Third highest
19:00	57.7%	Lowest (read from the chart)

- **Seasonal peaks are small.** December (60.7%), July (60.6%) and September (60.5%) run highest; April (59.2%) is lowest.
- **Midday and evening orders run slightly higher.** Orders placed at 12:00 and 20:00 are the most often late.
- **Process beats timing.** Timing moves the late rate by under 5 points; shipping mode moves it by about 60.

The notebook's day-of-week chart groups orders by day of the month (order_day), not by weekday, so it is left out. Recomputing it with .dt.dayofweek would fix it.

8 Predictive Model

A Random Forest model predicts late orders with 77% accuracy. When it flags an order as late, it is right 83% of the time.

Table 7. Model setup

Step	Detail
Target	Late_delivery_risk (1 = late, 0 = not late)
Features (10)	Shipping mode, order region, customer segment, department, category, payment type, scheduled shipping days, order month, order day, order hour
Encoding	Frequency encoding of the six categorical features
Split	Stratified 80/20 train-test split
Balancing	SMOTE on the training set only: on-time rows raised from 59,030 to 79,182 to match the 79,182 late rows
Model	Random Forest classifier (scikit-learn defaults)

Table 8. Test-set results (34,553 orders)

Metric	On time (class 0)	Late (class 1)	Weighted average
Precision	0.71	0.83	0.78
Recall	0.79	0.76	0.77
F1-score	0.75	0.79	0.77
Orders (support)	14,758	19,795	34,553
Accuracy			0.77

- **Reliable alerts.** With 0.83 precision on late orders, most flagged orders really are at risk, so alerts will not overwhelm the team with false alarms.
- **Good coverage.** The model catches 76% of late orders; the other 24% still slip through.
- **Weaker on on-time orders.** Precision for on-time orders is lower, at 0.71.

9 Recommendations

Fix the shipping-mode setup first, then deploy the alert model and clear payment holds. The remaining actions are medium and low priority.

Table 9. Recommended actions, ranked by impact

#	Action	Priority	Evidence
1	Audit First Class and Same Day carrier SLAs and delivery promises; until fixed, reprice these modes or reset the promise.	**Critical**	100% late; Second Class 79.8%
2	Deploy the model as a late-order alert at order confirmation, with early customer messages and priority handling.	**High**	77% accuracy; 0.83 precision
3	Speed up payment review and pending-order checks, starting in Central Africa.	**High**	95.0% and 70.2% late in Central Africa
4	Plan extra capacity ahead of July, September and December.	Medium	Highest months: 60.5% to 60.7%
5	Default eligible orders to Standard Class.	Medium	39.8% late, the best mode
6	Audit the hotspots: Central Asia and Pet Shop.	Medium	65.9% and 64.6% late
7	Review pricing and discounts on loss-making orders.	Low	18.7% of orders lose money (about $3.7M)
8	Retrain the model every quarter and test carrier, weather and warehouse data.	Low	Lift accuracy above 77%

10 Conclusion and Targets

Late delivery is the norm, not the exception, but its causes are clear and fixable.

60.09% of orders miss their scheduled date, and these orders carry $2.2M of profit. Shipping-mode setup is the dominant failure. Payment holds are a regional, secondary bottleneck. Timing effects are small, and no region or customer segment stands apart.

The Random Forest model already flags late orders with 77% accuracy, so it can be piloted now rather than later.

Table 10. Targets

Measure	Today	Target
Late delivery rate	60.09%	Below 30% within 12 months
First Class on-time rate	0%	Above 80%
Second Class on-time rate	20.2%	Above 60%
Model accuracy	77%	Above 82%
Loss-making orders	18.7%	Below 12%
Profit carried by late orders	$2.2M	Reduce by 40%

> **Next step**
>
> Start with the shipping-mode audit and the alert-model pilot. Together they address the largest driver of late orders and give the team early warning on the rest.

Appendix: Notes on the Numbers

- **Source.** Figures come from the notebook's printed outputs unless noted below.
- **Derived figures.** Net profit of about $3.8M is the sum of total profit across delay groups (Table 4). Losses of about $3.7M are $7.5M minus that net. The $22.57 and $21.67 averages are weighted from Table 4.
- **Read from charts.** Monthly values in Figure 5, other than the three annotated months, and the 19:00 hourly low (57.7%) are read from the notebook's charts.
- **Profit on late orders.** The notebook labels the $2.2M figure "Total Loss due to delays"; it is the total profit of orders with a delay above 0 days.
- **Same Day orders.** They are scheduled for 0 days and ship after about half a day, so the delay rule marks all of them late.
- **Model target.** The model predicts Late_delivery_risk (57.3% late in the data), not the 60.09% delay-based measure.
- **Actions and targets.** The actions in Table 9 and the target levels in Table 10 are adapted from the original report; their evidence and 'Today' values use the notebook's numbers.
- **Excluded.** The notebook's day-of-week chart uses day of the month, so it is not used in this report.